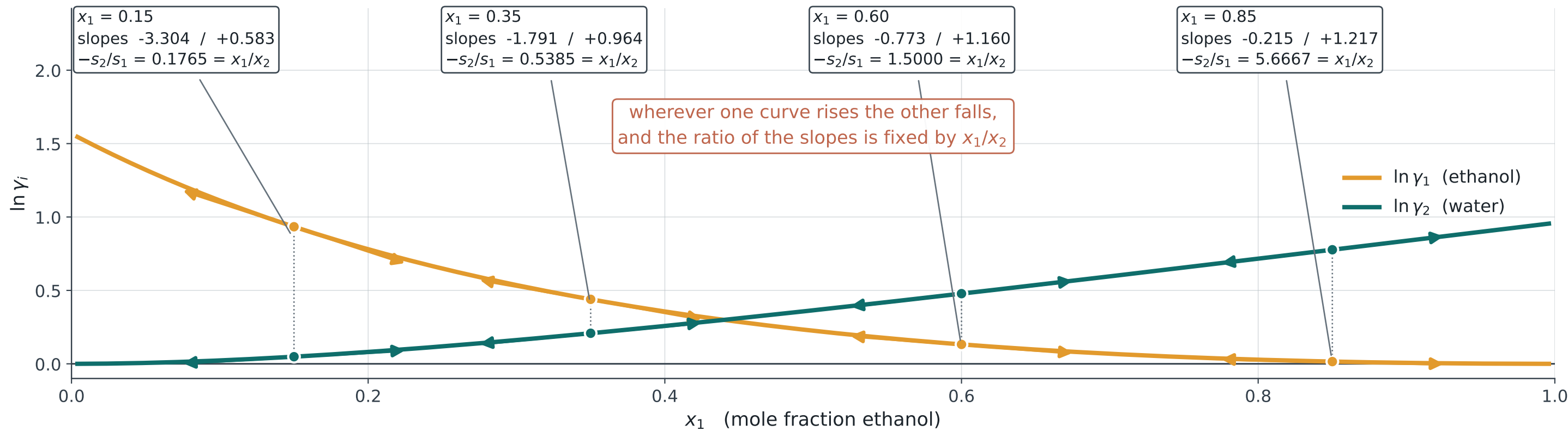
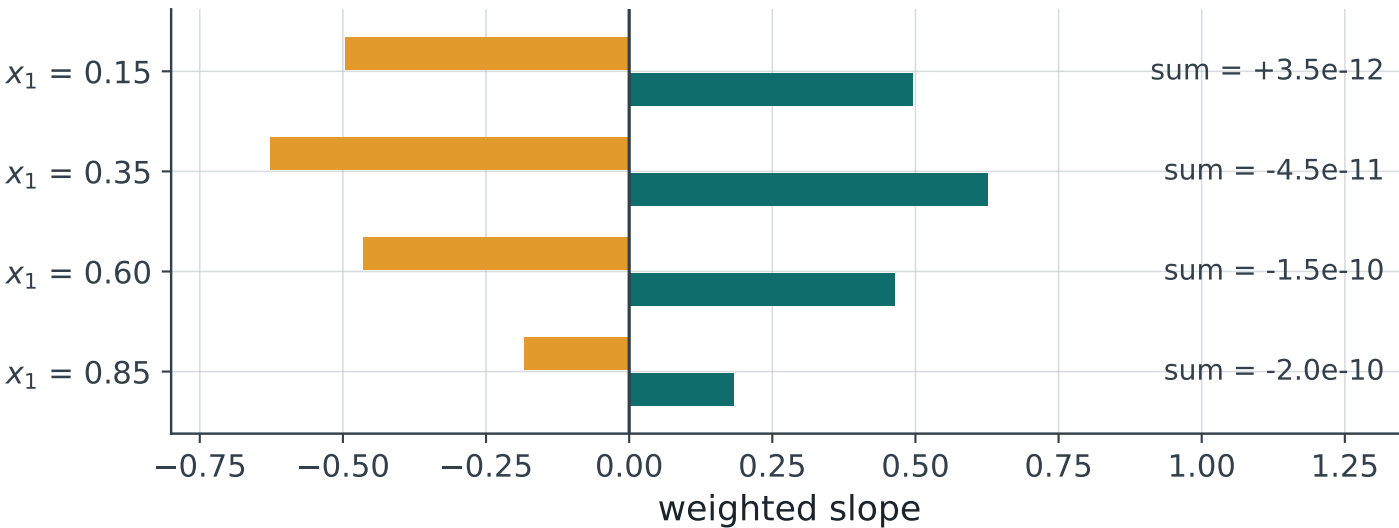


$x_1 d\ln \gamma_1/dx_1 + x_2 d\ln \gamma_2/dx_1 = 0$: the second activity coefficient is not free

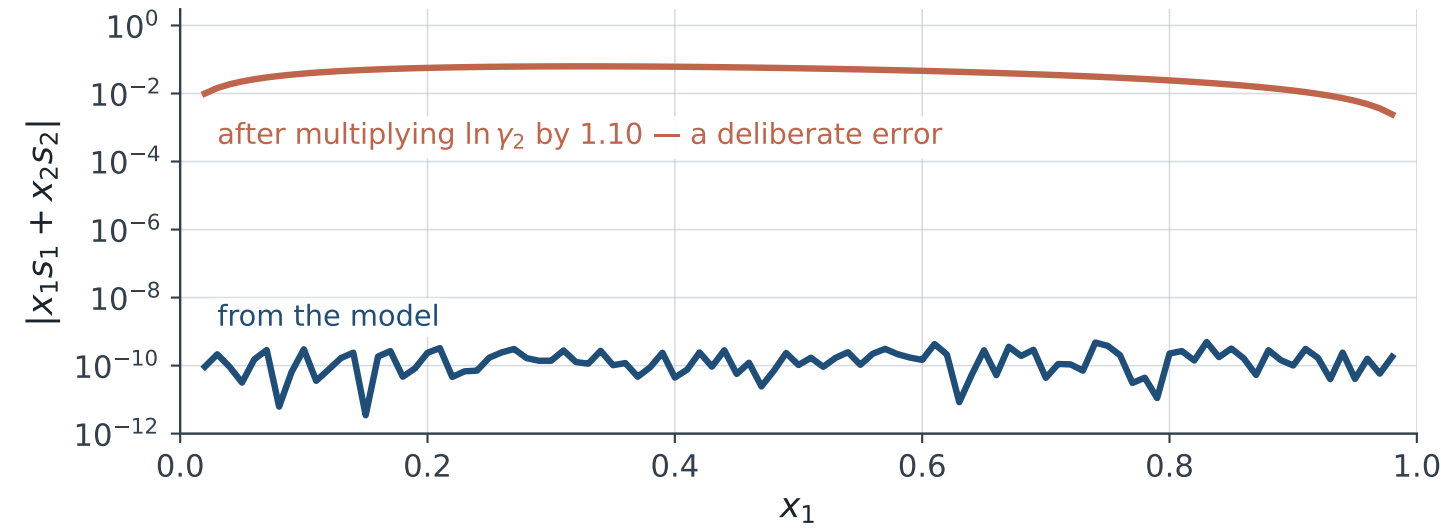
ethanol (1) + water (2) at 298.15 K, modified UNIFAC (Dortmund); $\gamma_1^\infty = 4.792$, $\gamma_2^\infty = 2.611$



$x_1 d\ln \gamma_1/dx_1$ (amber) and $x_2 d\ln \gamma_2/dx_1$ (teal):
equal and opposite at every composition, not approximately



the residual, on a log axis



Activity coefficients from `thermo.unifac.UNIFAC`, DOUFIP2006 table, DOUFSG subgroups; nothing fitted, nothing assumed. Slopes by central difference, $h = 1e-06$. The $\times 1.10$ curve in the right-hand panel is a DELIBERATE ARTIFICIAL ERROR introduced to show the constraint has content: it is not a property of any fluid.

Residual from the model: max $4.9e-10$ over 97 compositions — the truncation error of the difference, not physics. With the 10 % error in $\ln \gamma_2$: max 0.063, $1e+08$ times larger.